relevance: a grade of 0 marks irrelevant passages, while higher grades (i.e., 1, 2, and 3) mark more relevant passages. To account for this difference, we use the Normalized Discounted Cumulative Gain ($NDCG$) metric as the main for the TREC track results. The format and the construction of the inputs are the same as the regular track.

Similar to the regular track, we compute two versions of each metric: (1) the vanilla top-k version, where the model scores the passages that appear in the top-k passages from a BM25 retriever (e.g., the metric we refer to as $NDCG@10(topk = 30)$; (2) the boosted version, the model scores the passages in (1) *and* any passages that has an annotation in the relevance assessments provided (0, 1, 2, or 3) for that query even if they were not retrieved by BM25 (e.g., what we refer to as simply $NDCG@10$).

Data. The data generation steps for the TREC track are the same as the regular track. Both of the tracks use the same training dataset. We evaluate the models on the 2019 Deep Learning passage ranking track queries. This dataset has 200 queries, only 47 of which has associated judgments. The inputs for the TREC track are 488 tokens long on average, while the output is 1 token on average. We construct our prompts the same way for both of the tracks.

Pre-processing. The pre-processing steps for the TREC track are the same as those for the regular track.

Data Resources. We access the passage collection resource as in the regular track. We obtain the queries and the relevance assessments from `https://trec.nist.gov/data/deep2019.html` and `https://microsoft.github.io/msmarco/TREC-Deep-Learning-2019.html`.

B.3 Summarization

B.3.1 CNN/DM

Scenario Description. **CNN/DailyMail** (Hermann et al., 2015b) is a summarization scenario that contains news article from CNN and the DailyMail along with highlights which act as a summary for the article. As an example, an input for the scenario looks like: "(CNN) – An American woman died aboard a cruise ship that docked at Rio de Janeiro on Tuesday, the same ship on which 86 passengers previously fell ill, according to the state-run Brazilian news agency, Agencia Brasil. The American tourist died aboard the MS Veendam, owned by cruise operator Holland America. Federal Police told Agencia Brasil that forensic doctors were investigating her death. The ship's doctors told police that the woman was elderly and suffered from diabetes and hypertension, according the agency. The other passengers came down with diarrhea prior to her death during an earlier part of the trip, the ship's doctors said. The Veendam left New York 36 days ago for a South America tour.", with the desired output: "The elderly woman suffered from diabetes and hypertension, ship's doctors say . Previously, 86 passengers had fallen ill on the ship, Agencia Brasil says ." In general, for this scenario, inputs will contain a news article and outputs will contain 2-4 sentences that represent the highlights of the information in the news article. Performance for this scenario is measured using the ROUGE-2 metric (Lin, 2004), for overall summary quality. Faithfulness of generated summaries are measured using SummaC (Laban et al., 2022) and QAFactEval (Fabbri et al., 2022).

Data. CNN/DM is created by adapting the CNN and Dailymail articles in the Qusetion Answering dataset collected by Hermann et al. (2015b). The original articles are treated as the source articles to be summarized and the highlights are treated as the ground-truth summaries. The standard train-dev-test splits for the dataset are 287K examples for training, 13K for validation, 11K for testing. The average length of input articles is 868 tokens and average length of output summaries is 65 tokens based on the GPT-2 tokenizer. Since we are operating in the few-shot setting, we sampled training articles that were between 50 and 150 tokens in length, in order to be able to fit examples within the context token constraints.

Pre-processing. To process the dataset, we replace new lines with space and truncate the input article at 512 tokens, following standard practices in prior work (Zhang et al., 2020a).

Data Resources. We access the dataset at `https://huggingface.co/datasets/cnn_dailymail`. The dataset is made available through our benchmark. The dataset has a datasheet available at `https://huggingface.co/datasets/cnn_dailymail`. Further analyses of the dataset were provided by Jung et al. (2019), Bommasani & Cardie (2020) and Tejaswin et al. (2021).

B.3.2 XSUM

Scenario Description. **XSUM** (Narayan et al., 2018) is a summarization scenario that features articles from the British Broadcasting Corporation (BBC). As an example, an input for the scenario looks like: "Authorities said the incident took place on Sao Joao beach in Caparica, south-west of Lisbon. The National Maritime Authority said a middleaged man and a young girl died after they were unable to avoid the plane. [6 sentences with 139 words are abbreviated from here.] Other reports said the victims had been sunbathing when the plane made its emergency landing. [Another 4 sentences with 67 words are abbreviated from here.] Video footage from the scene carried by local broadcasters showed a small recreational plane parked on the sand, apparently intact and surrounded by beachgoers and emergency workers. [Last 2 sentences with 19 words are abbreviated.]". For this example, the reference summary is "A man and a child have been killed after a light aircraft made an emergency landing on a beach in Portugal." In general, for this scenario, inputs will contain news articles from BBC and outputs will contain a one sentence summary. Performance for this scenario is measured using the ROUGE-2 metric (Lin, 2004), for overall summary quality. Faithfulness of generated summaries are measured using SummaC (Laban et al., 2022) and QAFactEval (Fabbri et al., 2022). Since we are operating in the few-shot setting, we sampled training articles that were between 50 and 150 tokens in length, in order to be able to fit examples within the context token constraints.

Data. Narayan et al. (2018) created the XSUM dataset by harvesting online articles from the British Broadcasting Corporation (BBC). For each article, the first sentence is treated as the summary for the rest of the article text. The train-dev-test splits for the dataset are 204K examples for training, 11K for validation, and $11k$ for testing. Gehrmann et al. (2021) filter the original dataset using a faithfulness classifier and remove instances that are deemed unfaithful by the classifier. The filtered training set contains 23K training instances. We do a further manual inspection and whitelist a set of 50 instances, from which we sample the few-shot instances. The average length of input articles is 567 tokens and average length of output summaries is 25 tokens based on the GPT-2 tokenizer.

Pre-processing. To process the dataset, we replace new lines with space and truncate the input article at 512 tokens, following standard practices in prior work (Zhang et al., 2020a).

Data Resources. We access the dataset at `https://huggingface.co/datasets/GEM/xsum`. The dataset is made available through our benchmark. The dataset has a datasheet available at `https://huggingface.co/datasets/GEM/xsum`. Further analyses of the dataset were provided by Jung et al. (2019), Bommasani & Cardie (2020) and Tejaswin et al. (2021).

B.4 Sentiment analysis

B.4.1 IMDB

Scenario Description. **IMDB** (Maas et al., 2011) is a sentiment analysis scenario that features the positive/negative sentiment in movie review. As an example, an input for the scenario looks like: "Bromwell High is a cartoon comedy. It ran at the same time as some other programs about school life, such as "Teachers". My 35 years in the teaching profession lead me to believe that Bromwell High's satire is much closer to reality than is "Teachers". The scramble to survive financially, the insightful students who can see right through their pathetic teachers' pomp, the pettiness of the whole situation, all remind me of the schools I knew and their students. When I saw the episode in which a student repeatedly tried to burn down the school, I immediately recalled ..... at ..... High. A classic line: INSPECTOR: I'm here to sack one of your teachers. STUDENT: Welcome to Bromwell High. I expect that many adults of my age think that Bromwell High is far fetched. What a pity that it isn't!", with the desired output: "Positive". In general,

for this scenario, inputs will be a movie review and outputs will be Positive/Negative. Accuracy for this scenario is measured using whether the model generated the correct output, i.e. "Positive" or "Negative".

Data. Maas et al. (2011) created the dataset by creating positive and negative reviews, where a positive review has a score >= 7 out of 10 and a negative review has a score <= 4 out of 10. Reviews with neutral ratings are not included in the train/test sets. The train-test splits for the dataset are 25000-25000 with inputs on average being 1,309 tokens and outputs on average being 1 token based on the GPT-2 tokenizer.

Pre-processing. Apart from downloading the datasets and tokenizing the instances, no special pre-processing is performed for this dataset.

Data Resources. We access the dataset at `https://ai.stanford.edu/~amaas/data/sentiment/`. The dataset is made available through our benchmark. The dataset has no datasheet available.

B.5 Toxicity detection

B.5.1 CivilComments

Scenario Description. **CivilComments** (Borkan et al., 2019a) is a text classification scenario that features examples from the Civil Comments platform, a commenting plugin for independent news sites. As an example, an input for the scenario looks like: "Blame men. There's always an excuse to blame men for the failures of women. \n Question: Is the passage above toxic?" with the desired output: "True." In general, for this scenario, inputs will start with a comment (either toxic or non-toxic) followed by "Question: Is the passage above toxic?" and outputs will be either "True" or "False." Accuracy for this scenario is measured using exact match accuracy and quasi-exact match accuracy.

Data. Borkan et al. (2019a) created the dataset by scraping anonymized public comments from the Civil Comments platform. The train-dev-test splits for the dataset are 80% train, 20% test with inputs on average being 583.482 tokens and outputs on average being 1 token based on the GPT-2 tokenizer.

Pre-processing. We follow the pre-processing steps described in Koh et al. (2021). In following these steps, we also upload subsets that only comprise comments targeting a specific category, for each of the following categories: ["male", "female", "LGBTQ", "christian", "muslim", "other_religions", "black", "white"].

Data Resources. We access the dataset at `https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification/data`. The dataset is made available through our benchmark. The dataset has a datasheet available at `https://huggingface.co/datasets/civil_comments` (the HF dataset is taken directly from the Kaggle dataset).

B.6 Miscellaneous text classification

B.6.1 RAFT

Scenario Description. **RAFT** (Alex et al., 2021) is a text classification scenario that features tasks naturally occurring in real-world settings. As an example, an input for the scenario looks like: " In law, an overruling sentence is a statement that nullifies a previous case decision as a precedent, by a constitutionally valid statute or a decision by the same or higher ranking court which establishes a different rule on the point of law involved. Label the sentence based on whether it is overruling or not. Possible labels: 1. not overruling 2. overruling Sentence: consequently, it is overthrow. ", with the desired output: "overruling." In general, for this scenario, inputs will include a task-specific instruction followed by the target example to classify, and outputs will be one of the classes provided in the instructions. Accuracy for this scenario is measured using exact match.